Week 12 Notes

STAT 251 SECTION 01

This week finishes comparing two populations and then turns to categorical variables with more than two categories. It starts with one question about two proportions that trips people up: **when do we pool?** Then it extends the two-sample idea from proportions to **means**, first for two independent samples and then for matched pairs, and closes with the chi-square tests. The textbook sections are OpenStax 10.1 (https://openstax.org/books/introductory-statistics-2e/pages/10-1-two-population-means-with-unknown-standard-deviations) OpenStax 10.4 (https://openstax.org/books/introductory-statistics-2e/pages/10-4-matched-or-paired-samples) and OpenStax Chapter 11 (https://openstax.org/books/introductory-statistics-2e/pages/11-introduction).

When do we pool two proportions?

Every test and interval for two proportions uses the same estimate, $\hat{p}_1 - \hat{p}_2$, and an SE of the form

$$SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

We never know p_1 and p_2, so we have to put something in their place. The choice depends on what we are assuming:

Null hypothesis or goal	The SE we use	Pool?
$H_0 : p_1 = p_2$ (that is, $p_1 - p_2 = 0$)	$SE_0 = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$, $\hat{p} = \frac{X_1 + X_2}{n_1 + n_2}$	**pooled**
$H_0 : p_1 - p_2 = d_0$, with $d_0 \neq 0$	$SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	**unpooled**
a $100(1-\alpha)\%$ confidence interval for $p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) \pm Z_{1-\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	**unpooled**

Why the first row pools. A test works out what the data would look like *if* H_0 *were true*. If $H_0 : p_1 = p_2$ is true, there is only one proportion, p, and both samples estimate it. The best estimate of one p uses all the data, so we combine the samples: all the successes over all the observations, $\hat{p} = (X_1 + X_2)/(n_1 + n_2)$. This is the same reason the one-sample test uses p_0 in its SE: the SE is built from what H_0 says.

Why the other two rows do not. If H_0 says $p_1 - p_2 = d_0$ with $d_0 \neq 0$, the null itself says the two proportions are *different*, so there is no common p to estimate, and pooling the samples would mix two different populations. A confidence interval assumes no hypothesis at all, so again there is nothing to justify pooling. In both cases we estimate p_1 by $\hat{p}_1$ and p_2 by $\hat{p}_2$ separately.

The rule in one line: **pool only when H_0 says $p_1 = p_2$.**

Example: a confidence interval for the hives medications

In Comparing two proportions (lecturemenu11.html#comparing-two-proportions) (Week 11) we tested the two hives medications (OpenStax Example 10.8): 20 of 200 adults on medication A and 12 of 200 on medication B still had hives after 30 minutes. With the pooled SE_0, $z_{obs} = 1.47$ and the two-sided P-value was 0.1416, so at $\alpha = 0.01$ we did not reject $H_0 : p_A = p_B$.

Now **estimate** the difference with a 99% confidence interval. This time we do not assume H_0, so we do not pool:

1. $\hat{p}_A = 20/200 = 0.10$ and $\hat{p}_B = 12/200 = 0.06$, so $\hat{p}_A - \hat{p}_B = 0.04$.

2. The unpooled SE:

$$SE = \sqrt{\frac{0.10(0.90)}{200} + \frac{0.06(0.94)}{200}} = \sqrt{0.000450 + 0.000282} \approx 0.0271$$

   (The pooled SE_0 was 0.02713; the two are close here because the sample sizes are equal and the proportions are similar.)
3. For 99% confidence, $Z_{0.995} = 2.576$, so the margin of error is $2.576 \times 0.0271 \approx 0.0698$.
4. The interval is 0.04 ± 0.0698, or $(-0.0298,\ 0.1098)$.

We are 99% confident that the proportion of adults who still have hives is between 2.98 percentage points *lower* and 10.98 points *higher* on medication A than on medication B. The interval contains 0, which agrees with the test: the data are consistent with no difference.

Example: testing a difference that is not zero

In the Week 11 electric-vehicle practice problem (OpenStax Example 10.10), 135 of 1,343 younger adults and 12 of 232 older adults owned an EV, and we found that younger adults were more likely to own one. Suppose the marketing firm makes a stronger claim: **younger adults' ownership rate is more than 2 percentage points higher** than older adults'. (The 2-point claim is ours, not the book's.) Test it at $\alpha = 0.05$.

1. **Assumptions.** Two independent random samples. Each sample has at least 12 successes and at least 220 failures, so both are large enough.
2. **Hypotheses.** p_Y and p_O are the proportions of all younger and all older U.S. adults who own an EV. The claim is the alternative: $H_A : p_Y - p_O > 0.02$ (upper-tailed). The null is $H_0 : p_Y - p_O = 0.02$.
3. **Test statistic.** H_0 says the proportions differ by 0.02, so there is no common p and we **do not pool**. $\hat{p}_Y = 135/1343 \approx 0.1005$ and $\hat{p}_O = 12/232 \approx 0.0517$, a difference of 0.0488.

$$SE = \sqrt{\frac{0.1005(0.8995)}{1343} + \frac{0.0517(0.9483)}{232}} \approx 0.0167, \qquad z_{obs} = \frac{0.0488 - 0.02}{0.0167} \approx 1.72$$

4. **P-value.** $P(Z \geq 1.72 \mid H_0 \text{ true}) = 1 - 0.9573 = 0.0427$.
5. **Decision.** $0.0427 < 0.05$: **reject** H_0. The data support the stronger claim, but only just.

The only change from an ordinary two-proportion test is in step 3: the hypothesized difference, 0.02, replaces the 0 in the numerator, and the SE is unpooled.

Comparing two independent means

(OpenStax Section 10.1 (https://openstax.org/books/introductory-statistics-2e/pages/10-1-two-population-means-with-unknown-standard-deviations).) So far we have compared two populations on a **categorical** response (a proportion). Now the response is **quantitative**, and we compare the populations' **means**: the average time girls and boys spend playing sports, or the average final-exam score in online and face-to-face sections. As with proportions, two sample means will almost never be equal, even when the population means are. The test asks whether the difference we see is bigger than ordinary sampling variation would produce.

The data are **two independent simple random samples**, one from each population:

Population	Population mean	Population SD	Sample size	Sample mean	Sample SD
1	μ_1	σ_1	n_1	$\bar{x}_1$	s_1

Population	Population mean	Population SD	Sample size	Sample mean	Sample SD
2	μ_2	σ_2	n_2	$\bar{x}_2$	s_2

The parameter is the **difference in population means**, $\mu_1 - \mu_2$, and its estimate is the difference in sample means, $\bar{x}_1 - \bar{x}_2$.

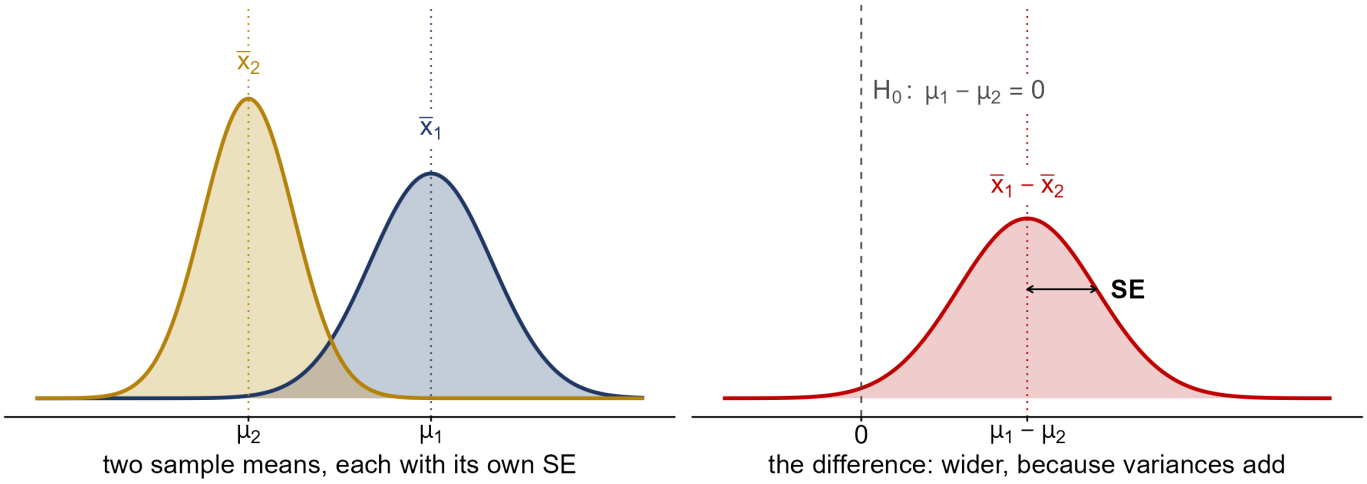
The sampling distribution of $\bar{x}_1 - \bar{x}_2$

- **Centre $\mu_1 - \mu_2$.** Each $\bar{x}$ is centred at its own μ, so on average the difference of the estimates is the difference of the parameters.
- **Spread.** From the one-sample case, $\bar{x}_1$ has variance σ_1^2/n_1 and $\bar{x}_2$ has variance σ_2^2/n_2. Because the samples are **independent, the variances add**, just as they did for proportions:

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

- **Shape.** Normal if both populations are normal. If they are not, the CLT makes it approximately normal when both samples are large (the course's rule is both $n \geq 30$).

Putting these together, $\bar{x}_1 - \bar{x}_2 \sim N\left(\mu_1 - \mu_2,\ \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right)$. The picture shows why the variances add. Each sample mean wobbles around its own μ, and their difference picks up both wobbles, so it is more spread out than either mean on its own.



Its distribution, in practice. We do not know σ_1 and σ_2, so the SE uses s_1 and s_2 instead (next section). The difference itself stays approximately normal. What changes is the **standardized** difference: dividing by an *estimated* SE adds uncertainty, and the result follows, approximately, a t distribution:

$$\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \ \dot{\sim}\ t\Big(\min(n_1 - 1,\ n_2 - 1)\Big)$$

($\dot{\sim}$ means "approximately follows"). Both the t shape and the df are approximations, as the next section explains.

The test statistic

In practice σ_1 and σ_2 are unknown, so, as in the one-sample t-test, we replace them with the sample standard deviations:

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Under H_0, $\mu_1 - \mu_2 = 0$, so the test statistic is

$$t_{obs} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Estimating two standard deviations adds uncertainty, so t_{obs} is compared with a t distribution, not the standard normal. Unlike the one-sample test, though, **the t distribution here is an approximation**. With two different SDs in the SE, t_{obs} does not follow any t distribution exactly, so both the curve and its degrees of freedom are approximate:

- **The Welch df (what software uses).** The best approximation, due to Welch (the textbook calls it the Aspin-Welch formula), is the df that makes a t curve match t_{obs} most closely. Writing $v_1 = s_1^2/n_1$ and $v_2 = s_2^2/n_2$ (each sample's share of the squared SE),

  $$df = \frac{(v_1 + v_2)^2}{\frac{v_1^2}{n_1 - 1} + \frac{v_2^2}{n_2 - 1}}$$

  It is usually not a whole number, and it always lies between $\min(n_1 - 1, n_2 - 1)$ and $n_1 + n_2 - 2$. For the sports example below, $v_1 = 0.866^2/9 = 0.0833$ and $v_2 = 1.00^2/16 = 0.0625$, so

  $$df = \frac{(0.0833 + 0.0625)^2}{0.0833^2/8 + 0.0625^2/15} \approx 18.85$$

- **In practice, by hand:** $df \approx \min(n_1 - 1,\ n_2 - 1)$, the smaller sample size minus one. This is a quick, rough approximation to the Welch df. It is never larger than the Welch df, so its t curve has heavier tails and our P-value comes out a little **larger** than software's. That makes the hand method **conservative**: if it rejects H_0, software will too. For the sports example it gives 8 instead of 18.85.

So, if H_0 is true, t_{obs} approximately follows a t distribution, and by hand we use $\min(n_1 - 1, n_2 - 1)$ degrees of freedom.

The hypotheses and the P-value

$$H_0 : \mu_1 = \mu_2, \qquad \text{which is the same as} \qquad H_0 : \mu_1 - \mu_2 = 0$$

Alternative hypothesis	Tail	In words	P-value
$H_A : \mu_1 - \mu_2 > 0$	upper-tailed	population 1 has the larger mean	$P(T \geq t_{obs} \mid H_0 \text{ true})$
$H_A : \mu_1 - \mu_2 < 0$	lower-tailed	population 1 has the smaller mean	$P(T \leq t_{obs} \mid H_0 \text{ true})$
$H_A : \mu_1 - \mu_2 \neq 0$	two-sided	the means differ, in either direction	$2 \times P(T \geq \lvert t_{obs} \rvert \mid H_0 \text{ true})$

With the t-table (misc/t_table.html) we usually cannot find the P-value exactly; we **bracket** it. Find the row for our df, find the two entries that $|t_{obs}|$ falls between, and read their upper-tail areas from the top of the table. For a two-sided test, double both areas.

Assumptions

1. **Two independent random samples**, one from each population (or two groups formed by random assignment). Nothing in one sample is paired with anything in the other.
2. **Normal populations, or large samples.** If both samples have at least 30 observations, the CLT takes care of the shape. With smaller samples, both populations should be roughly normal (look for strong skew or outliers in the data).

The five steps

1. **Assumptions.** Independent random samples; normal populations or both $n \geq 30$.
2. **Hypotheses.** Define μ_1 and μ_2 in words, then $H_0 : \mu_1 - \mu_2 = 0$ and the alternative.
3. **Test statistic.** The SE, t_{obs}, and $df = \min(n_1 - 1, n_2 - 1)$.
4. **P-value.** Bracket it with the t-table, using the tail(s) that H_A points to.
5. **Decision.** Reject H_0 if the P-value $< \alpha$; state the conclusion in the context of the problem.

A confidence interval for $\mu_1 - \mu_2$

To estimate the difference rather than test it, use the same SE and df:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{1-\alpha/2} \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \qquad df = \min(n_1 - 1, n_2 - 1)$$

As always, an interval that leaves out 0 matches a two-sided test that rejects $H_0 : \mu_1 - \mu_2 = 0$.

To pool or not (means)

There is a second version of the two-sample t-test, the **pooled** t-test. If we are willing to assume the two populations have the **same standard deviation**, $\sigma_1 = \sigma_2 = \sigma$, then both samples estimate the same σ, and we combine them into the **pooled standard deviation**:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

The SE becomes $s_p\sqrt{1/n_1 + 1/n_2}$, and the test statistic follows a t distribution with exactly $n_1 + n_2 - 2$ degrees of freedom.

Test or interval	The SE (or interval) we use	df	Pool?
t-test of $H_0 : \mu_1 = \mu_2$, assuming $\sigma_1 = \sigma_2$	$SE = s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$	$n_1 + n_2 - 2$	**pooled**
t-test of $H_0 : \mu_1 = \mu_2$, σ_1, σ_2 possibly different (our default)	$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$\approx \min(n_1 - 1, n_2 - 1)$ by hand; Welch in software	**unpooled**
a $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$, assuming $\sigma_1 = \sigma_2$	$(\bar{x}_1 - \bar{x}_2) \pm t_{1-\alpha/2}$ $s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$	$n_1 + n_2 - 2$	**pooled**

Test or interval	The SE (or interval) we use	df	Pool?
a $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$, σ_1, σ_2 possibly different (our default)	$(\bar{x}_1 - \bar{x}_2)$ $\pm t_{1-\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$\approx \min(n_1 - 1, n_2 - 1)$ by hand; Welch in software	**unpooled**

Notice the difference from proportions. There, the *hypothesis* decided whether to pool: the test of $H_0 : p_1 = p_2$ pools, and the confidence interval does not. For means, the test and the interval always make the **same** choice, because what decides it is an *assumption about the SDs*, $\sigma_1 = \sigma_2$, not anything H_0 says.

We do not pool for means. The textbook is blunt about it: right after the df formula it says "Notice that the sample variances $(s_1)^2$ and $(s_2)^2$ are not pooled. (If the question comes up, do not pool the variances.)" (OpenStax p. 513, and again in Example 10.1). The book does not give a reason. Here it is:

- The pooled test is correct **only if** $\sigma_1 = \sigma_2$, and we almost never know that. Testing for equal SDs first does not really help, because that test has little power with small samples, which is exactly when it matters.
- If the SDs differ, especially when the sample sizes differ too, the pooled test's real Type I error rate can be far from α: it rejects a true H_0 more often (or less often) than we think.
- The unpooled (Welch) test is valid whether or not the SDs are equal, and when they *are* equal it loses very little. So it is the safe default, and it is what most software does unless you ask otherwise.

Compare this with proportions. There we *must* pool for the test of $H_0 : p_1 = p_2$, because for a proportion the SD is determined by p itself, $\sqrt{p(1-p)}$. A null that says the proportions are equal automatically says the SDs are equal too. $H_0 : \mu_1 = \mu_2$ says nothing at all about σ_1 and σ_2, so it gives us no reason to pool.

Advantages and disadvantages of pooling

- **When it is reasonable.** The sample SDs are similar (a common rule of thumb: the larger is less than about twice the smaller), and the sample sizes are similar too. Pooling is also natural in a randomized experiment where the treatment is expected to shift the mean but not the spread.
- **The advantage.** If $\sigma_1 = \sigma_2$ really holds, the pooled t statistic follows a t distribution *exactly*, with $n_1 + n_2 - 2$ degrees of freedom, more than either the Welch df or our $\min(n_1 - 1, n_2 - 1)$. More df means thinner tails, smaller P-values and **more power**. The worked example after the sports test below shows this.
- **The disadvantage.** If the SDs differ, especially when the sample sizes differ too, the pooled test's real Type I error rate is not α. A simulation makes this concrete. Suppose H_0 is true (both means equal), but sample 1 has $n_1 = 10$ from a population with $\sigma_1 = 3$ and sample 2 has $n_2 = 40$ with $\sigma_2 = 1$. Repeating that study 20,000 times:

```
set.seed(251)
rej <- replicate(20000, {
  x <- rnorm(10, mean = 0, sd = 3)     # small sample, large SD
  y <- rnorm(40, mean = 0, sd = 1)     # large sample, small SD; H0 is true
  c(pooled = t.test(x, y, var.equal = TRUE)$p.value < 0.05,
    welch  = t.test(x, y)$p.value < 0.05)
})
rowMeans(rej)     # how often each 5% test rejects a TRUE null
```

```
## pooled  welch
## 0.2578 0.0497
```

The "5%" pooled test rejects the true null about **26%** of the time: more than five times as many false alarms as promised. Welch's test stays at about 5%. (Swap which sample has the larger SD and the pooled test goes the other way, almost never rejecting, which costs power.) * **Our rule.** We default to the unpooled test. When pooling is justified, it gains a little power; when it is not, it can badly mislead, and we rarely know which case we are in.

The textbook does use s_p in one place, in Cohen's d, a measure of effect size (p. 518), but never for the test.

Example: girls, boys and sports

From OpenStax Introductory Statistics 2e, Section 10.1, Example 10.1 (https://openstax.org/books/introductory-statistics-2e/pages/10-1-two-population-means-with-unknown-standard-deviations):

> The average amount of time boys and girls aged seven to 11 spend playing sports each day is believed to be the same. A study is done and data are collected, resulting in the data in the table below. Each population has a normal distribution.
>
> | | Sample size | Average number of hours playing sports per day | Sample standard deviation |
> |---|---|---|---|
> | Girls | 9 | 2 | 0.866 |
> | Boys | 16 | 3.2 | 1.00 |
>
> Is there a difference in the mean amount of time boys and girls aged seven to 11 play sports each day? Test at the 5% level of significance.

1. **Assumptions.** Two independent random samples. The samples are small (9 and 16), but both populations are normal, so the t-test works.
2. **Hypotheses.** Let μ_g and μ_b be the mean daily hours of sports for all girls and all boys aged 7 to 11. "Believed to be the same" is the null, and the question has no direction, so

$$H_0 : \mu_g - \mu_b = 0 \qquad H_A : \mu_g - \mu_b \neq 0 \quad \text{(two-sided)}$$

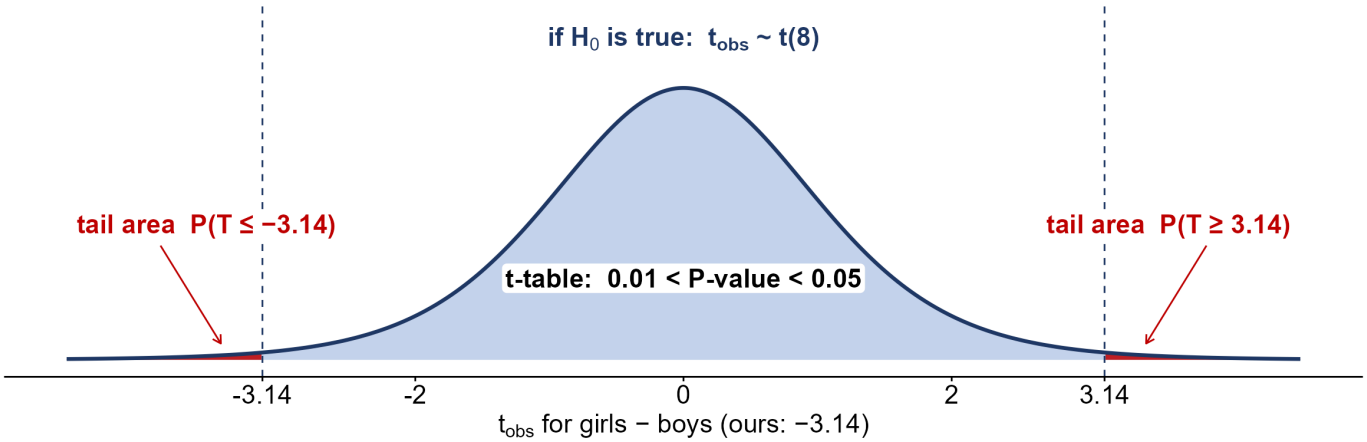
3. **Test statistic.**

$$SE = \sqrt{\frac{0.866^2}{9} + \frac{1.00^2}{16}} = \sqrt{0.0833 + 0.0625} \approx 0.3819, \qquad t_{obs} = \frac{2 - 3.2}{0.3819} \approx -3.14$$

   with $df = \min(9 - 1, 16 - 1) = 8$.
4. **P-value.** In row 8 of the t-table, $2.306 < 3.14 < 3.355$. Those entries have upper-tail areas 0.025 and 0.005, so $0.005 < P(T \leq -3.14) < 0.025$. Doubling for two sides,

$$0.01 < \text{P-value} < 0.05$$

   Software, with df = 8, gives 0.0138.
5. **Decision.** The P-value is below $\alpha = 0.05$: **reject** H_0. The data give convincing evidence that girls and boys aged 7 to 11 differ in the mean time they spend playing sports each day; the sample suggests boys play more.



Why the book's P-value is smaller. The book reports a P-value of 0.0054. It uses the Welch df, about 18.85, and our hand method uses df = 8. The t curve with 8 df has heavier tails, so our P-value is larger. The conclusion is the same. This is the conservative trade-off: when the hand method rejects H_0, so does software.

Example: math classes at two colleges

From OpenStax Introductory Statistics 2e, Section 10.1, Example 10.2 (https://openstax.org/books/introductory-statistics-2e/pages/10-1-two-population-means-with-unknown-standard-deviations):

> A study is done by a community group in two neighboring colleges to determine which one graduates students with more math classes. College A samples 11 graduates. Their average is four math classes with a standard deviation of 1.5 math classes. College B samples nine graduates. Their average is 3.5 math classes with a standard deviation of one math class. The community group believes that a student who graduates from college A **has taken more math classes**, on the average. Both populations have a normal distribution. Test at a 1% significance level.

1. **Assumptions.** Two independent random samples of graduates; both populations are normal, so the small samples are fine. The SDs are unknown, so it is a t-test of two means.
2. **Hypotheses.** μ_A and μ_B are the mean number of math classes taken by all graduates of colleges A and B. The group's belief is the alternative:

$$H_0 : \mu_A - \mu_B = 0 \qquad H_A : \mu_A - \mu_B > 0 \quad \text{(upper-tailed)}$$

   (The book writes the null as $H_0 : \mu_A \leq \mu_B$, a composite null; as with composite hypotheses in Week 11, we test it at the boundary, $\mu_A - \mu_B = 0$.)
3. **Test statistic.**

$$SE = \sqrt{\frac{1.5^2}{11} + \frac{1^2}{9}} = \sqrt{0.2045 + 0.1111} \approx 0.5618, \qquad t_{obs} = \frac{4 - 3.5}{0.5618} \approx 0.89, \qquad df = \min(10, 8) = 8$$

4. **P-value.** In row 8, $0.889 < 0.89 < 1.108$, with upper-tail areas 0.20 and 0.15, so $0.15 < \text{P-value} < 0.20$. Software, with df = 8, gives 0.1997 (the book, with the Welch df, 0.1928).
5. **Decision.** The P-value is far above $\alpha = 0.01$: **do not reject** H_0. There is not enough evidence that graduates of college A take more math classes, on average, than graduates of college B.

Example: online and face-to-face final exams

From OpenStax Introductory Statistics 2e, Section 10.1, Example 10.3 (https://openstax.org/books/introductory-statistics-2e/pages/10-1-two-population-means-with-unknown-standard-deviations):

> A professor at a large community college wanted to determine whether there is a difference in the means of final exam scores between students who took his statistics course online and the students who took his face-to-face statistics class. He believed that the mean of the final exam scores for the online class would be lower than that of the face-to-face class. Was the professor correct? The randomly selected 30 final exam scores from each group are listed below. Is the mean of the Final Exam scores of the online class lower than the mean of the Final Exam scores of the face-to-face class? Test at a 5% significance level.

Class	Score 1	Score 2	Score 3	Score 4	Score 5	Score 6	Score 7	Score 8	Score 9	Score 10
Online	67.6	41.2	85.3	55.9	82.4	91.2	73.5	94.1	64.7	64.7
	70.6	38.2	61.8	88.2	70.6	58.8	91.2	73.5	82.4	35.5
	94.1	88.2	64.7	55.9	88.2	97.1	85.3	61.8	79.4	79.4
Face-to-face	77.9	95.3	81.2	74.1	98.8	88.2	85.9	92.9	87.1	88.2
	69.4	57.6	69.4	67.1	97.6	85.9	88.2	91.8	78.8	71.8
	98.8	61.2	92.9	90.6	97.6	100.0	95.3	83.5	92.9	89.4

1. **Assumptions.** Two independent random samples of 30 scores each. Both $n \geq 30$, so we do not need the populations to be normal.
2. **Hypotheses.** μ_{on} and μ_f are the mean final-exam scores of all the professor's online and face-to-face students. His belief is the alternative:

$$H_0 : \mu_{on} - \mu_f = 0 \qquad H_A : \mu_{on} - \mu_f < 0 \quad \text{(lower-tailed)}$$

3. **Test statistic.** From the data (a calculator or R does this step): online $\bar{x} = 72.85$, $s = 16.92$; face-to-face $\bar{x} = 84.98$, $s = 11.71$.

$$SE = \sqrt{\frac{16.92^2}{30} + \frac{11.71^2}{30}} \approx 3.7568, \qquad t_{obs} = \frac{72.85 - 84.98}{3.7568} \approx -3.23, \qquad df = \min(29, 29) = 29$$

4. **P-value.** In row 29, $3.038 < 3.23 < 3.659$, with upper-tail areas 0.0025 and 0.0005, so $0.0005 < \text{P-value} < 0.0025$. Software, with df = 29, gives 0.0015 (the book, with the Welch df, 0.0011).
5. **Decision.** The P-value is below 0.05: **reject** H_0. The professor was right: the data give strong evidence that the mean final-exam score is lower for his online students than for his face-to-face students. (This is an observational comparison, so it does not show that the online format *causes* the lower scores; students chose their section.)

Example: the 2024 NCAA women's basketball final

On Sunday, April 7, 2024, South Carolina and Iowa met in the NCAA women's basketball championship. The table gives the points each team scored in its last 15 games before the final (from the 2024 STAT 251 notes).

Game	South Carolina	Opponent	Iowa	Opponent
1	78	NC State	71	UConn
2	70	Oregon State	94	LSU
3	79	Indiana	89	Colorado
4	88	UNC	64	West Virginia
5	91	Presbyterian	91	Holy Cross
6	79	LSU	94	Nebraska
7	74	Tennessee	95	Michigan
8	79	Texas A&M	95	Penn State
9	76	Tennessee	93	Ohio State
10	98	Arkansas	108	Minnesota
11	103	Kentucky	101	Illinois
12	72	Alabama	69	Indiana
13	70	Georgia	106	Michigan
14	66	Tennessee	79	Nebraska
15	83	UConn	111	Penn State

Question: At $\alpha = 0.05$, is there a difference between the two teams in mean points scored per game?

1. **Assumptions.** We treat each team's 15 games as a sample of its games that season; the two samples are independent (different games). With $n = 15$, we need the points per game to be roughly normal, and the data show no strong skew or extreme outliers. (Recent games are not a true random sample, so this is an approximation.)
2. **Hypotheses.** μ_I and μ_S are the mean points per game for Iowa and South Carolina.

$$H_0 : \mu_I - \mu_S = 0 \qquad H_A : \mu_I - \mu_S \neq 0 \quad \text{(two-sided)}$$

3. **Test statistic.** Iowa: $\bar{x} = 90.67$, $s = 14.21$. South Carolina: $\bar{x} = 80.40$, $s = 10.57$.

$$SE = \sqrt{\frac{14.21^2}{15} + \frac{10.57^2}{15}} \approx 4.5727, \qquad t_{obs} = \frac{90.67 - 80.40}{4.5727} \approx 2.25, \qquad df = 14$$

4. **P-value.** In row 14, $2.145 < 2.25 < 2.977$, with upper-tail areas 0.025 and 0.005; doubling, $0.01 < \text{P-value} < 0.05$. Software gives 0.041.
5. **Decision.** $\text{P-value} < 0.05$: **reject** H_0, narrowly. Iowa scored significantly more points per game. (South Carolina won the final anyway, 87–75: a test about averages says nothing certain about any one game.)

Practice: who keeps workers longer?

From OpenStax Introductory Statistics 2e, Section 10.1, Try It 10.2 (https://openstax.org/books/introductory-statistics-2e/pages/10-1-two-population-means-with-unknown-standard-deviations) (the book gives no solution; the worked solution is ours):

A study is done to determine if Company A retains its workers longer than Company B. Company A samples 15 workers, and their average time with the company is five years with a standard deviation of 1.2. Company B samples 20 workers, and their average time with the company is 4.5 years with a standard deviation of 0.8. The populations are normally distributed.

a. Are the population standard deviations known?
b. Conduct an appropriate hypothesis test. At the 5% significance level, what is your conclusion?

▶ Solution

Example: pooling for girls and boys

Go back to girls, boys and sports (Example 10.1). The sample SDs are close, 0.866 and 1.00 (the larger is only 1.15 times the smaller), so assuming the two populations have the same SD, $\sigma_g = \sigma_b$, is reasonable. What changes if we pool?

1. **The pooled SD.** With $n_g = 9$ and $n_b = 16$,

$$s_p = \sqrt{\frac{(9-1)(0.866^2) + (16-1)(1.00^2)}{9+16-2}} = \sqrt{\frac{6.000 + 15}{23}} \approx 0.9555$$

   It lies between the two sample SDs, closer to the boys' because their sample is larger.
2. **The pooled SE and test statistic.**

$$SE = s_p\sqrt{\frac{1}{9} + \frac{1}{16}} = 0.9555 \times 0.4167 \approx 0.3981, \qquad t_{obs} = \frac{2 - 3.2}{0.3981} \approx -3.01, \qquad df = 9 + 16 - 2 = 23$$

3. **P-value.** In row 23, $2.807 < 3.01 < 3.104$, with upper-tail areas 0.005 and 0.0025; doubling, $0.005 < \text{P-value} < 0.01$. Software gives 0.0062.
4. **Compare.** The unpooled test gave $t_{obs} = -3.14$ on 8 df and $0.01 < \text{P-value} < 0.05$. Pooling made the SE slightly *larger* here (0.3981 against 0.3819), but it nearly tripled the degrees of freedom, and the extra df shrink the P-value. That is the payoff of pooling, *when the equal-SD assumption is right*: a more powerful test. Both tests reject H_0 at the 5% level, so the conclusion is the same.

A non-parametric alternative: the rank-sum test

The two independent samples t-test needs normal populations or large samples. When the samples are small and the data are clearly skewed, have outliers, or are only ranks or ordered categories (such as ratings from 1 to 7), the t-test's P-value can be wrong. The **Wilcoxon rank-sum test** (also called the Mann-Whitney test) is the **non-parametric alternative**, just as the sign test (lecturemenu11.html#the-sign-test) was for one sample or for pairs:

- It needs two independent random samples, but **no normality assumption**.
- It **ranks** all $n_1 + n_2$ observations together, from smallest to largest, and adds up the ranks belonging to each sample. If the two populations have the same centre, the ranks should be well mixed and each sample's rank sum should be close to what we would expect from its size. A rank sum that is much too large or too small is evidence that one population tends to give larger values.
- Because it uses ranks rather than the values, one extreme outlier counts as just "the largest value", so it cannot dominate the test the way it can dominate a mean and an SD.
- The price: when the populations really are normal, the t-test is a little more powerful.

For matched pairs, the non-parametric partner of the matched-pairs t-test is the **Wilcoxon signed-rank test**, a stronger cousin of the sign test that also uses the sizes of the differences.

Both tests, with worked examples, are in the extra notes under **Resources** → **Extras** → **Non-parametric Tests**: Non-parametric tests for two samples (misc/Non-Parametric-Tests-For-Two-Samples.html).

Matched pairs

(OpenStax Section 10.4 (https://openstax.org/books/introductory-statistics-2e/pages/10-4-matched-or-paired-samples).) Recall that two samples can be **dependent**: each observation in one sample is paired with one in the other. The most common designs are

- **the same subjects measured twice**, before and after a treatment (blood pressure before and after a drug), or under two conditions (each student's dominant and weaker hand);
- **natural or deliberate pairs**, such as twins, or two plots of land side by side, with one member of each pair getting each treatment.

The two-sample t-test above would be **wrong** for these data, because it assumes that the samples are independent. Instead, we use the pairing.

From two samples to one

For each pair, compute the **difference** $d = x_1 - x_2$ (for example, after − before). The n differences become the data, and the two-sample problem turns into a **one-sample** problem: the parameter is the **mean difference** μ_d in the population of pairs, and we test it with the one-sample t-test from Testing a population mean (lecturemenu10.html#testing-a-population-mean) (Week 10).

Symbol	Meaning
d	the difference within a pair, $x_1 - x_2$ (always say which way you subtract)
n	the number of pairs (not the number of measurements)
μ_d	the population mean difference: the parameter
$\bar{d}$	the mean of the n sample differences: the estimate of μ_d
s_d	the standard deviation of the n sample differences

The test

- **Hypotheses.** $H_0 : \mu_d = 0$, no difference within pairs on average, against $H_A : \mu_d > 0$ (upper-tailed), $H_A : \mu_d < 0$ (lower-tailed) or $H_A : \mu_d \neq 0$ (two-sided). Which direction is "upper" depends on the order of subtraction, so fix d first.
- **Test statistic.**

$$t_{obs} = \frac{\bar{d} - 0}{s_d/\sqrt{n}}, \qquad df = n - 1$$

  If H_0 is true, t_{obs} follows a t distribution with $n - 1$ degrees of freedom.
- **P-value.** From the t-table, exactly as for the one-sample test.
- **Confidence interval.** $\bar{d} \pm t_{1-\alpha/2} \times s_d/\sqrt{n}$, with $df = n - 1$.

Assumptions.

1. The pairs are a random sample (or the treatment was randomly assigned within each pair), and the pairs are independent of one another.
2. The **differences** are approximately normal, or there are at least 30 pairs. The two original measurements do not each need to be normal; only the differences matter. With few pairs, look at the differences for skew and outliers.

Why pair?

Pairing removes the variation **between subjects**. People differ a lot in how much pain they feel, how far they can throw, or how high their blood pressure is. In an independent-samples design, all of that person-to-person variation goes into s_1 and s_2, and so into the SE. When each subject is compared with themself, those differences cancel out of d, s_d is smaller, and the test is more powerful: it can detect a smaller effect with the same number of subjects. The hypnotism example below shows this with numbers.

The sign test (Week 11) is also a matched-pairs test. It uses only the **signs** of the differences, so it needs no normality assumption; the matched-pairs t-test uses their **sizes**, so it is more powerful when the differences really are roughly normal.

Example: does hypnotism reduce pain?

From OpenStax Introductory Statistics 2e, Section 10.4, Example 10.11 (https://openstax.org/books/introductory-statistics-2e/pages/10-4-matched-or-paired-samples):

> A study was conducted to investigate the effectiveness of hypnotism in reducing pain. Results for randomly selected subjects are shown in the table below. A lower score indicates less pain. The "before" value is matched to an "after" value and the differences are calculated. The differences have a normal distribution. Are the sensory measurements, on average, lower after hypnotism? Test at a 5% significance level.

Subject	A	B	C	D	E	F	G	H
Before	6.6	6.5	9.0	10.3	11.3	8.1	6.3	11.6
After	6.8	2.4	7.4	8.5	8.1	6.1	3.4	2.0
d = after − before	0.2	−4.1	−1.6	−1.8	−3.2	−2.0	−2.9	−9.6

1. **Assumptions.** Matched pairs: the same 8 subjects were measured before and after hypnotism. The subjects were randomly selected, and the differences are normal (given), so the small sample is fine.
2. **Hypotheses.** Let d = after − before, and μ_d the mean difference for all such subjects. Less pain after hypnotism means *negative* differences, so

$$H_0 : \mu_d = 0 \qquad H_A : \mu_d < 0 \quad \text{(lower-tailed)}$$

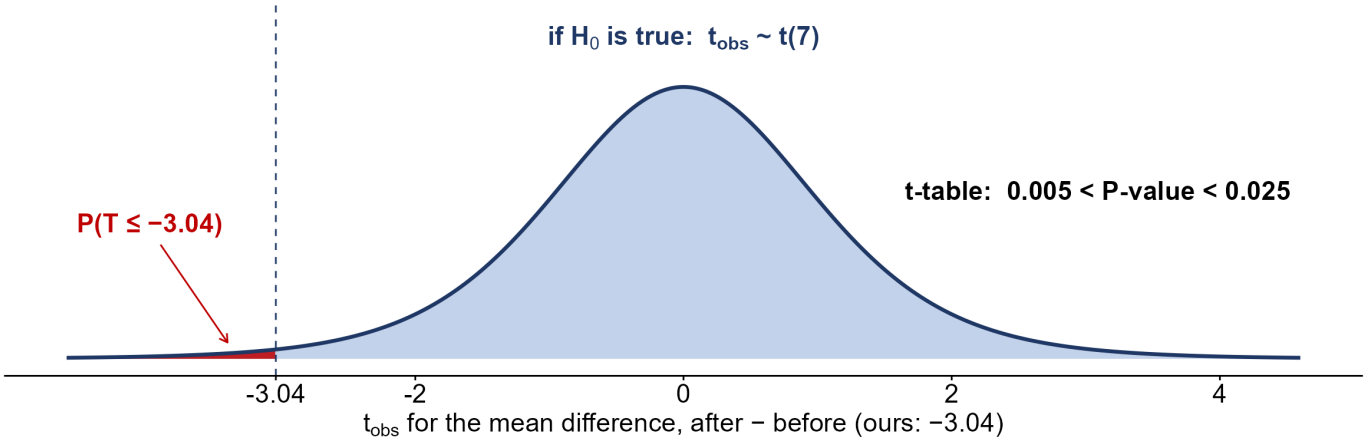
   (The book writes $H_0 : \mu_d \geq 0$; we test it at the boundary, 0.)
3. **Test statistic.** The 8 differences add to -25.0, so $\bar{d} = -25.0/8 = -3.125$, and their standard deviation is $s_d \approx 2.91$.

$$SE = \frac{s_d}{\sqrt{n}} = \frac{2.91}{\sqrt{8}} \approx 1.0288, \qquad t_{obs} = \frac{-3.125 - 0}{1.0288} \approx -3.04, \qquad df = 8 - 1 = 7$$

4. **P-value.** In row 7, $2.365 < 3.04 < 3.499$, with upper-tail areas 0.025 and 0.005. By symmetry the lower tail beyond -3.04 has the same area, so

$$0.005 < P(T \leq -3.04 \mid H_0 \text{ true}) < 0.025$$

   Software gives 0.0094.
5. **Decision.** The P-value is below 0.05: **reject** H_0. The data give convincing evidence that pain scores are lower after hypnotism, on average.



What if we had ignored the pairing? Treat the before and after scores as two independent samples of 8: before $\bar{x} = 8.71$, $s = 2.18$; after $\bar{x} = 5.59$, $s = 2.61$. The unpooled SE is

$$\sqrt{\frac{2.18^2}{8} + \frac{2.61^2}{8}} \approx 1.2023, \qquad t_{obs} = \frac{-3.125}{1.2023} \approx -2.60$$

The estimate is the same, -3.125, but the SE is larger (1.2023 against 1.0288), so t_{obs} is smaller. Subjects who start with a high pain score tend to stay relatively high, and the independent-samples SE counts that person-to-person spread as noise. The paired analysis removes it. Ignoring the pairing is not just less powerful; it is the **wrong test**, because the two samples are not independent.

Example: the softball strength class

From OpenStax Introductory Statistics 2e, Section 10.4, Example 10.12 (https://openstax.org/books/introductory-statistics-2e/pages/10-4-matched-or-paired-samples):

> A college softball coach was interested in whether the college's strength development class increased their players' maximum lift (in pounds). Four players participated in the study. The amount of weight they could each lift was recorded before they took the strength development class. After completing the class, the amount of weight they could each lift was again measured. The data are as follows:
>
> | Weight (in pounds) | Player 1 | Player 2 | Player 3 | Player 4 |
> |---|---|---|---|---|
> | Amount of weight lifted prior to the class | 205 | 241 | 338 | 368 |
> | Amount of weight lifted after the class | 295 | 252 | 330 | 360 |
>
> The coach wants to know if the strength development class makes the players stronger, on average.

The book does not give α; we use 0.05, as the book's solution does.

1. **Assumptions.** Matched pairs (the same 4 players, before and after). The book assumes the differences are normal; we come back to that below.
2. **Hypotheses.** d = after — before, so "stronger" means positive differences: $H_0 : \mu_d = 0$ and $H_A : \mu_d > 0$ (upper-tailed).
3. **Test statistic.** The differences are $90, 11, -8, -8$. So $\bar{d} = 85/4 = 21.25$ and $s_d \approx 46.70$.

$$SE = \frac{46.70}{\sqrt{4}} = 23.35, \qquad t_{obs} = \frac{21.25}{23.35} \approx 0.91, \qquad df = 3$$

4. **P-value.** In row 3, $0.765 < 0.91 < 0.978$, with upper-tail areas 0.25 and 0.20, so $0.20 < \text{P-value} < 0.25$. Software (and the book) give 0.215.
5. **Decision. Do not reject H_0.** There is not enough evidence that the class makes players stronger, on average.

A caution. Look at the differences: three of the four players lifted about the same or *less* after the class, and one player's 90-pound gain makes $\bar{d}$ positive. With 4 pairs and one possible outlier, the assumption that the differences are normal is very doubtful, and so is any conclusion. The coach needs more players.

Example: shot-put with each hand

From OpenStax Introductory Statistics 2e, Section 10.4, Example 10.13 (https://openstax.org/books/introductory-statistics-2e/pages/10-4-matched-or-paired-samples):

> Seven eighth graders at Kennedy Middle School measured how far they could push the shot-put with their dominant (writing) hand and their weaker (non-writing) hand. They thought that they could push equal distances with either hand. The data were collected and recorded in the table below.
>
> | Distance (in feet) using | Student 1 | Student 2 | Student 3 | Student 4 | Student 5 | Student 6 | Student 7 |
> |---|---|---|---|---|---|---|---|
> | Dominant hand | 30 | 26 | 34 | 17 | 19 | 26 | 20 |
> | Weaker hand | 28 | 14 | 27 | 18 | 17 | 26 | 16 |
>
> Conduct a hypothesis test to determine whether the mean difference in distances between the children's dominant versus weaker hands is significant.

The book assumes the differences are normal and uses $\alpha = 0.05$.

1. **Assumptions.** Matched pairs: each student used both hands. The differences are assumed normal.
2. **Hypotheses.** d = dominant — weaker. "Equal distances with either hand" is the null, and the question has no direction: $H_0 : \mu_d = 0$ and $H_A : \mu_d \neq 0$ (two-sided).
3. **Test statistic.** The differences are $2, 12, 7, -1, 2, 0, 4$. So $\bar{d} = 26/7 \approx 3.71$ and $s_d \approx 4.50$.

$$SE = \frac{4.50}{\sqrt{7}} \approx 1.7008, \qquad t_{obs} = \frac{3.71}{1.7008} \approx 2.18, \qquad df = 6$$

4. **P-value.** In row 6, $1.943 < 2.18 < 2.447$, with upper-tail areas 0.05 and 0.025; doubling, $0.05 < \text{P-value} < 0.10$. Software gives 0.072 (the book: 0.0716).
5. **Decision.** The P-value is above 0.05: **do not reject** H_0. There is not enough evidence of a mean difference between the students' dominant and weaker hands. Six of the seven students pushed at least as far with their dominant hand, so a larger study might well find a difference.

Example: who made the decision? (a t-test instead of the sign test)

In the Week 11 notes we used the **sign test** on 17 couples' ratings of who had more say in a major financial decision (1 = the wife, 7 = the husband), and found a two-sided P-value of 0.0574: not significant at the 5% level. What does the matched-pairs t-test say about the same data?

The differences d = husband − wife add to 23, so $\bar{d} = 23/17 \approx 1.35$, and $s_d \approx 1.77$.

$$SE = \frac{1.77}{\sqrt{17}} \approx 0.4293, \qquad t_{obs} = \frac{1.35}{0.4293} \approx 3.15, \qquad df = 16$$

In row 16, $2.921 < 3.15 < 3.252$, with upper-tail areas 0.005 and 0.0025; doubling, $0.005 < \text{P-value} < 0.01$. Software gives 0.006. The t-test **rejects** H_0, and the sign test did not.

Which one should we believe? The t-test uses the sizes of the differences, so it has more power, **if** its assumption holds. But these are crude 1-to-7 ratings from only 17 couples, and it is hard to believe the differences come from a normal population. The sign test makes no such assumption. That is why we used it there. When the assumption behind a test is doubtful, a more powerful-looking P-value is not more trustworthy. Here the honest summary is "suggestive but not conclusive".

Practice: throwing with each hand

From OpenStax Introductory Statistics 2e, Section 10.4, Try It 10.13 (https://openstax.org/books/introductory-statistics-2e/pages/10-4-matched-or-paired-samples) (the book gives no solution; the worked solution is ours, assuming the differences are normal):

> Five ball players think they can throw the same distance with their dominant hand (throwing) and off-hand (catching hand). The data were collected and recorded in the table below. Conduct a hypothesis test to determine whether the mean difference in distances between the dominant and off-hand is significant. Test at the 5% level.
>
> | | Player 1 | Player 2 | Player 3 | Player 4 | Player 5 |
> |---|---|---|---|---|---|
> | Dominant hand | 120 | 111 | 135 | 140 | 125 |
> | Off-hand | 105 | 109 | 98 | 111 | 99 |

▶ Solution

Choosing a test

The first question is always **what kind of response** (categorical or quantitative), and the second is **how the samples are related** (one sample, two independent samples, or matched pairs).

Parameter	Design	Response	Test statistic	Distribution	Confidence interval
p	one sample	categorical	$z_{obs} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$	$N(0,1)$	$\hat{p} \pm Z_{1-\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$
$p_1 - p_2$	two independent samples	categorical	$z_{obs} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$ (pooled $\hat{p}$)	$N(0,1)$	$(\hat{p}_1 - \hat{p}_2) \pm Z_{1-\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$
μ	one sample	quantitative	$t_{obs} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$t(n-1)$	$\bar{x} \pm t_{1-\alpha/2}\, s/\sqrt{n}$
$\mu_1 - \mu_2$	two independent samples	quantitative	$t_{obs} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$	$t(\min(n_1-1, n_2-1))$	$(\bar{x}_1 - \bar{x}_2) \pm t_{1-\alpha/2}\sqrt{s_1^2/n_1 + s_2^2/n_2}$
μ_d	matched pairs	quantitative	$t_{obs} = \frac{\bar{d}}{s_d/\sqrt{n}}$	$t(n-1)$	$\bar{d} \pm t_{1-\alpha/2}\, s_d/\sqrt{n}$

Some quick checks:

- **The blood pressure of 30 patients before and after a drug:** the same patients twice, so matched pairs.
- **Mean nightly sleep for random samples of 40 first-years and 40 seniors:** two separate samples and a quantitative response, so the two independent samples t-test.
- **The share of 200 dog owners and 200 cat owners who exercise daily:** two independent samples and a yes/no response, so the two-proportion z-test.

Summary of the two-sample tests

Every two-sample procedure from Lectures 26–28, side by side. "Large" checks use the course's rule of 15 (for proportions and the sign test) or 30 (for means).

Scenario	Hypotheses	SE	Test statistic	$100(1-\alpha)\%$ CI	Distribution if H_0 is true	Assumptions
Two proportions, test of no difference (two independent samples, yes/no response)	$H_0: p_1 - p_2 = 0$ $H_A: p_1 - p_2 >, <, \neq 0$	$\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$, $\hat{p} = \frac{X_1+X_2}{n_1+n_2}$ (pooled)	$z_{obs} = \frac{\hat{p}_1 - \hat{p}_2}{SE}$	(use the unpooled interval in the next row)	$N(0,1)$	two independent random samples; $n_1\hat{p}, n_1(1-\hat{p}), n_2\hat{p}, n_2(1-\hat{p})$ all ≥ 15 (pooled $\hat{p}$)
Two proportions, test of a non-zero difference	$H_0: p_1 - p_2 = d_0$ $H_A: p_1 - p_2 >, <, \neq d_0$	$\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ (unpooled)	$z_{obs} = \frac{(\hat{p}_1 - \hat{p}_2) - d_0}{SE}$	$(\hat{p}_1 - \hat{p}_2) \pm Z_{1-\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	$N(0,1)$	two independent random samples; at least 15 successes and 15 failures in each sample

Scenario	Hypotheses	SE	Test statistic	CI	Distribution if is true	Assumptions
Two means, unpooled (our default) (two independent samples, quantitative response)	$H_0: \mu_1 - \mu_2 = 0$ $H_A: \mu_1 - \mu_2 >, <, \neq 0$	$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$t_{obs} = \frac{\bar{x}_1 - \bar{x}_2}{SE}$	$(\bar{x}_1 - \bar{x}_2) \pm t_{1-\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	approximately t, $df \approx \min(n_1 - 1, n_2 - 1)$ (Welch df in software)	two independent random samples; both populations normal, or both $n \geq 30$
Two means, pooled	$H_0: \mu_1 - \mu_2 = 0$ $H_A: \mu_1 - \mu_2 >, <, \neq 0$	$s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$, $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$	$t_{obs} = \frac{\bar{x}_1 - \bar{x}_2}{SE}$	$(\bar{x}_1 - \bar{x}_2) \pm t_{1-\alpha/2}$ $s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$	$t(n_1 + n_2 - 2)$, exactly if $\sigma_1 = \sigma_2$	as unpooled, **plus** $\sigma_1 = \sigma_2$ (similar sample SDs and sizes)
Matched pairs, mean difference (the same subjects twice, or matched pairs)	$H_0: \mu_d = 0$ $H_A: \mu_d >, <, \neq 0$	$\frac{s_d}{\sqrt{n}}$, n = number of pairs	$t_{obs} = \frac{\bar{d} - 0}{s_d/\sqrt{n}}$	$\bar{d} \pm t_{1-\alpha/2}\frac{s_d}{\sqrt{n}}$	$t(n-1)$	a random sample of pairs; the differences normal, or $n \geq 30$ pairs
Matched pairs, sign test (Week 11) (non-parametric)	$H_0: M_D = 0$ (median difference) $H_A: M_D >, <, \neq 0$	$SE(S) = \frac{\sqrt{n}}{2}$, n = non-zero differences	S = number of $+$ differences; large n: $z_{obs} = \frac{S - n/2}{\sqrt{n}/2}$	not covered	$S \sim$ Binomial $(n, 0.5)$; large n: $N(0,1)$	a random sample of pairs; no normality needed; $n/2 \geq 15$ for the z version
Two independent samples, rank-sum test (non-parametric)	the two populations have the same centre, against one tending to give larger values	see the extra notes	the rank sum of one sample	not covered	rank-sum tables or software	two independent random samples; no normality needed

More on the two non-parametric tests, with worked examples, is under **Resources** → **Extras** → **Non-parametric Tests**: Non-parametric tests for two samples (misc/Non-Parametric-Tests-For-Two-Samples.html).

Every test so far: which SE?

Every test statistic has the same shape, (estimate − null value) / SE. What changes from test to test is the SE, and for two independent samples there are two versions. We **pool** only when H_0 says the two proportions are equal ($H_0: p_1 = p_2$), or when two means come from populations with equal SDs ($\sigma_1 = \sigma_2$) and that is a reasonable assumption. Otherwise we use

the **unpooled** SE. Note that the one-proportion test uses the null value p_0 in its SE, because H_0 tells us what p is.

Test	Samples	Test statistic	SE, unpooled	SE, pooled
one proportion z	one sample	$z_{obs} = \frac{\hat{p}-p_0}{SE}$	$\sqrt{\frac{p_0(1-p_0)}{n}}$	—
one mean t	one sample	$t_{obs} = \frac{\bar{x}-\mu_0}{SE}$	$\frac{s}{\sqrt{n}}$	—
two proportions z	two independent samples	$z_{obs} = \frac{(\hat{p}_1-\hat{p}_2)-d_0}{SE}$	$\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ (for $H_0: p_1-p_2=d_0\neq 0$, and every CI)	$\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}, \hat{p} = \frac{X_1+X_2}{n_1+n_2}$ (only for $H_0: p_1=p_2$)
two means t	two independent samples	$t_{obs} = \frac{\bar{x}_1-\bar{x}_2}{SE}$	$\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}$ (our default)	$s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$ (only if $\sigma_1=\sigma_2$ is reasonable)
matched pairs t	dependent (pairs)	$t_{obs} = \frac{\bar{d}-0}{SE}$	$\frac{s_d}{\sqrt{n}}$, n = number of pairs	—

The two non-parametric tests sit beside these. The **sign test** (Week 11) is for pairs, and the **rank-sum test** is for two independent samples.

Categorical data with more than two categories

(OpenStax Chapter 11 (https://openstax.org/books/introductory-statistics-2e/pages/11-introduction).) So far, every categorical response has had **two** outcomes, such as yes or no, or success or failure. We summarized it with one proportion p, or compared two groups with $p_1 - p_2$. Many categorical variables have more than two categories, though: a survey answer on a scale, a blood type, a political party, a number of hours grouped into ranges. The questions we ask about them change too:

- **One variable:** does its distribution match a claimed set of proportions? For example, are all four answers to a survey question equally common? This is the **goodness-of-fit test**, later in these notes.
- **Two variables:** are they related? For example, does the number of hours a person volunteers depend on what type of volunteer they are? This is the **test of independence**, later in these notes.

A test of one proportion cannot answer either question. It looks at one category at a time, while the claim covers every category at once, and the category proportions are tied together because they must add to 1. The new tests compare whole tables of counts at once, and all of them use the **chi-square distribution**.

Data: teens and phones

Recall the Pew Research Center survey from Week 2. It asked 743 U.S. teens aged 13 to 17, *"Are you losing focus in class by checking your cell phone?"*

X = a teen's answer: Never, Rarely, Sometimes or Often. $n = 743$.

Answer	Never	Rarely	Sometimes	Often	Total
Count	289	216	178	60	743
Proportion	0.39	0.29	0.24	0.08	1.00

This is a **one-way table**, which gives the count in each category of one categorical variable.

Two-way tables

A **two-way table**, also called a **contingency table**, gives the count for every combination of two categorical variables. The categories of one variable are the rows and those of the other are the columns. From a two-way table we can read two kinds of distribution:

- The **marginal distribution** of a variable is that variable on its own. It is read from the row or column *totals* in the margins of the table, and is usually written as proportions of the grand total.
- A **conditional distribution** is one variable *within a single row (or column)* of the other: each count in that row divided by the row total.

If the two variables are unrelated, every row has about the same conditional distribution, and so each row looks like the column totals. When the rows differ a lot, the variables are associated.

> **Example 11.6 (OpenStax).** In a volunteer group, adults 21 and older volunteer from one to nine hours each week to spend time with a disabled senior citizen. The program recruits among community college students, four-year college students, and nonstudents. The table is a sample of the adult volunteers and the number of hours they volunteer per week. Is the number of hours volunteered independent of the type of volunteer?

OpenStax Section 11.3, Example 11.6 (https://openstax.org/books/introductory-statistics-2e/pages/11-3-test-of-independence). X = type of volunteer, Y = hours volunteered per week, $n = 839$.

Type of volunteer	1–3 hours	4–6 hours	7–9 hours	Row total
Community college students	111	96	48	**255**
Four-year college students	96	133	61	**290**
Nonstudents	91	150	53	**294**
Column total	**298**	**379**	**162**	**839**

Do the rows look alike? Divide each row by its row total to get the conditional distribution of hours for each type of volunteer, and divide the column totals by 839 to get the marginal distribution:

Type of volunteer	1–3 hours	4–6 hours	7–9 hours
Community college students	0.435	0.376	0.188
Four-year college students	0.331	0.459	0.210
Nonstudents	0.310	0.510	0.180
All volunteers (marginal)	0.355	0.452	0.193

For example, $111/255 = 0.435$ of the community college volunteers work 1–3 hours, compared with $91/294 = 0.310$ of the nonstudents and $298/839 = 0.355$ of all volunteers. Community college students lean toward fewer hours and nonstudents toward 4–6 hours. Whether differences this large could be ordinary sampling variation is exactly what the test of independence, later in these notes, decides.

Observed and expected counts

Every chi-square test compares two sets of counts:

- the **observed count** O is the count in a category (or a cell of a two-way table) in our data;
- the **expected count** E is the count we would expect *if* H_0 *were true*. For one variable whose null hypothesis gives category i the proportion p_i, the expected count is

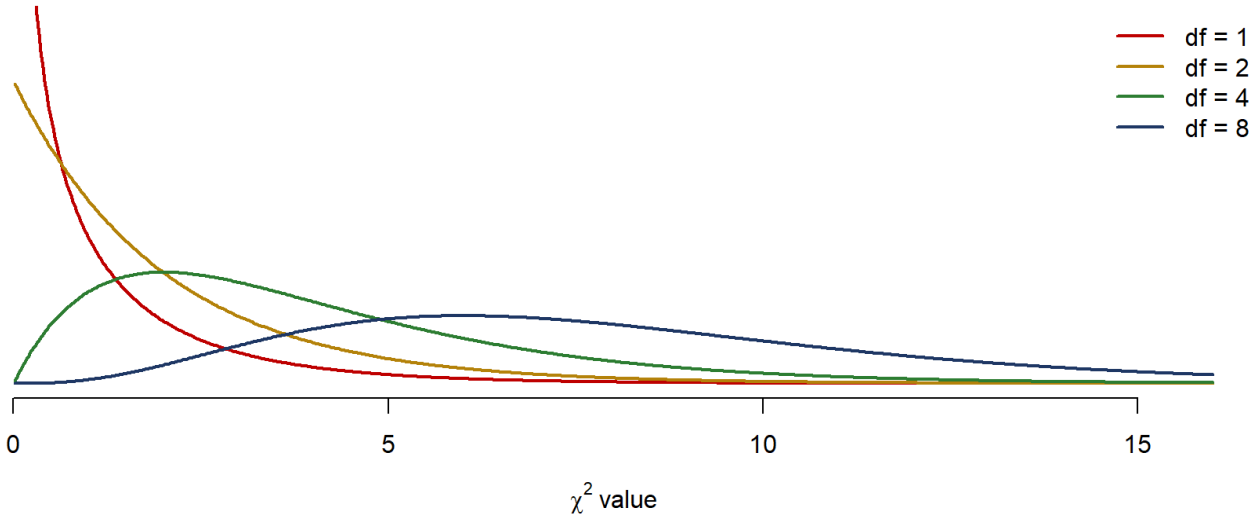
$$E = n\,p_i.$$

If the observed counts are close to the expected counts, the data agree with H_0. If they are far apart, the data are evidence against it.

The chi-square distribution

(OpenStax Section 11.1 (https://openstax.org/books/introductory-statistics-2e/pages/11-1-facts-about-the-chi-square-distribution).) The **chi-square distribution**, written $\chi^2(df)$, is a family of curves, one for each number of degrees of freedom, just as with the t distribution. Its facts:

- A chi-square value is **never negative**. The curve starts at 0 and is **skewed right**.
- Its **mean is** df and its standard deviation is $\sqrt{2\,df}$.
- As df grows, the curve moves right, spreads out, and becomes more symmetric. For very large df it is close to a normal curve.
- For one categorical variable with k categories, the degrees of freedom are $df = k - 1$.



The chi-square statistic

The **chi-square statistic** adds up, over every category, how far each observed count is from its expected count:

$$X^2 = \sum \frac{(O - E)^2}{E}.$$

- Each term is squared, so X^2 is never negative, and $X^2 = 0$ only when every observed count equals its expected count.
- Dividing by E scales each gap. A gap of 10 matters more when we expected 20 than when we expected 2000.
- A **large** X^2 means the counts are far from what H_0 predicts, which is evidence against H_0. So chi-square tests are always **right-tailed**.
- If H_0 is true and every expected count is at least 5, X^2 approximately follows a chi-square distribution. For one variable with k categories that is $\chi^2(k-1)$.

Example: are the four answers equally common? Someone claims that the four answers to the Pew question are equally common among U.S. teens, so $H_0 : p_1 = p_2 = p_3 = p_4 = 0.25$, where $p_1, \dots, p_4$ are the population proportions answering Never, Rarely, Sometimes and Often.

a. The expected count for each answer is $E = 743 \times 0.25 = 185.75$. It does not need to be a whole number, because it is an average.

Answer	Observed O	Expected E	$O - E$	$(O-E)^2/E$
Never	289	185.75	103.25	57.39
Rarely	216	185.75	30.25	4.93
Sometimes	178	185.75	-7.75	0.32
Often	60	185.75	-125.75	85.13

b. $X^2 = 57.39 + 4.93 + 0.32 + 85.13 = 147.77$.

c. $df = 4 - 1 = 3$. A $\chi^2(3)$ curve has mean 3 and standard deviation $\sqrt{6} = 2.45$, so 147.77 is far out in its right tail. The four answers are clearly not equally common: far more teens said "Never" and far fewer said "Often" than the claim predicts. The goodness-of-fit test below turns this into a full test with a P-value.

Three chi-square tests

The statistic X^2 is the basis of three tests, one for each kind of question we can ask of categorical data. All three use the same statistic,

$$X^2 = \sum \frac{(O-E)^2}{E},$$

and all three are right-tailed. What changes from test to test is the *question*, which sets the null hypothesis, and that in turn sets how we find the expected counts E and the degrees of freedom.

(Reading: OpenStax Sections 11.2 to 11.5 (https://openstax.org/books/introductory-statistics-2e/pages/11-2-goodness-of-fit-test). Chi-square P-values in these notes are bracketed with the course chi-square table (misc/chisq_table.html), and the exact value from software is given beside each bracket.)

The chi-square goodness-of-fit test

The setup

A **goodness-of-fit test** asks whether **one categorical variable** follows a claimed distribution: a claimed proportion for each of its k categories. It is the categorical version of a one-sample test.

- **The parameters.** $p_1, p_2, \dots, p_k$, the proportions of the *population* in each category. They always add to 1.
- **The hypotheses.** The null hypothesis gives every category its claimed proportion, written $p_{1,0}, p_{2,0}, \dots, p_{k,0}$ (they add to 1 too):

$$H_0 : p_1 = p_{1,0},\ p_2 = p_{2,0},\ \dots,\ p_k = p_{k,0}$$

$$H_A : \text{at least one } p_i \text{ is not its claimed value}$$

  In words: H_0 says the population *fits* the claimed distribution, and H_A says it does not. There is no "upper" or "lower" version of H_A: the counts can miss the claim in many different ways at once.
- **The expected counts.** If H_0 is true, category i should hold about a fraction $p_{i,0}$ of the sample:

$$E_i = n\,p_{i,0}$$

  The expected counts add to n, just as the observed counts do.
- **The test statistic.**

$$X^2_{obs} = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

- **The null distribution.** If H_0 is true (and the conditions below hold), X^2_{obs} approximately follows a chi-square distribution with

$$df = k - 1.$$

  Why $k-1$ and not k? The observed counts must add to n. Once you know $k-1$ of them, the last one is fixed, so only $k-1$ counts are free to vary. **The df comes from the number of categories, not the sample size.**
- **The P-value.** Only a *large* X^2_{obs} is evidence against H_0, so the P-value is always the upper tail:

$$\text{P-value} = P(\chi^2 \geq X^2_{obs} \mid H_0 \text{ true})$$

- **The conditions.** (1) The data are a random sample, with each observation counted in exactly one category. (2) **Every expected count is at least 5.** The chi-square curve is only an approximation, and it is a poor one when some E_i are small. Note that the condition is on the *expected* counts, not the observed counts.

Reading the chi-square table. Go to the row for your df. Each column heading is an upper-tail area. Find the two entries your X^2_{obs} falls between; the P-value is between their two column headings. If X^2_{obs} is smaller than the first entry, the P-value is more than 0.25; if it is larger than the last entry, the P-value is less than 0.0005. For example, the row for $df = 4$ starts

Upper tail	0.25	0.2	0.15	0.125	0.1	0.075	0.05	0.025	0.005	0.0025	5e-04
$df = 4$	5.385	5.989	6.745	7.214	7.779	8.496	9.488	11.143	14.860	16.424	19.997

The five steps

1. **Assumptions.** A random sample; every expected count $E_i = n\,p_{i,0} \geq 5$.
2. **Hypotheses.** Name the categories and the claimed proportions; state H_0 and H_A in symbols or words.
3. **Test statistic.** Make the table of O, E, $O - E$ and $(O-E)^2/E$; add the last column to get X^2_{obs}; $df = k - 1$.
4. **P-value.** Bracket $P(\chi^2 \geq X^2_{obs} \mid H_0 \text{ true})$ with the chi-square table.
5. **Decision.** Reject H_0 if the P-value $< \alpha$, and state the conclusion in context.

Example: are the four answers equally common?

The chi-square statistic section left the Pew teens-and-phones example at $X^2 = 147.77$ with $df = 3$. Here are all five steps.

1. **Assumptions.** Pew surveyed a random sample of 743 U.S. teens, and each expected count is 185.75, far above 5.
2. **Hypotheses.** With $p_1, \ldots, p_4$ the population proportions answering Never, Rarely, Sometimes and Often,

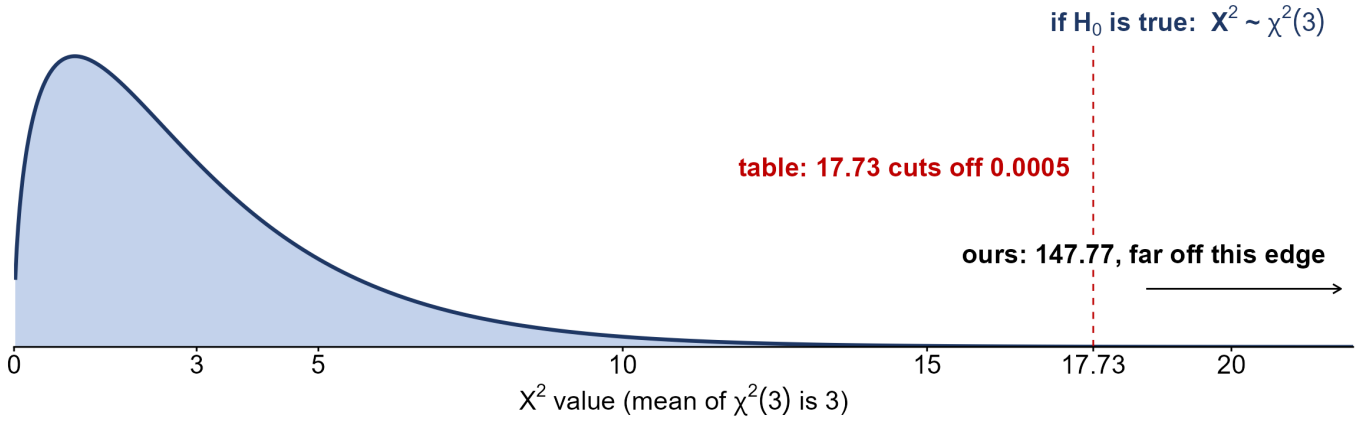
$$H_0 : p_1 = p_2 = p_3 = p_4 = 0.25 \qquad H_A : \text{at least one } p_i \neq 0.25$$

3. **Test statistic.** From above, $X^2_{obs} = 147.77$ with $df = 4 - 1 = 3$.
4. **P-value.** The largest entry in the $df = 3$ row is 17.730, in the 0.0005 column, and 147.77 is far beyond it:

$$P(\chi^2 \geq 147.77 \mid H_0 \text{ true}) < 0.0005$$

   (Software gives a number with 31 zeros after the decimal point.)

5. **Decision. Reject H_0** at any usual α. The four answers are not equally common among U.S. teens.



A rejection says only that the claim is wrong. To say *how* it is wrong, look back at which terms of X^2 were large: "Often" (85.13) and "Never" (57.39). Far fewer teens than claimed say they often lose focus, and far more say they never do.

Example: which day are employees absent?

From OpenStax Introductory Statistics 2e, Section 11.2, Example 11.2 (https://openstax.org/books/introductory-statistics-2e/pages/11-2-goodness-of-fit-test):

> Employers want to know which days of the week employees are absent in a five-day work week. Most employers would like to believe that employees are absent equally during the week. Suppose a random sample of 60 managers were asked on which day of the week they had the highest number of employee absences. The results were distributed as in the table below. For the population of employees, do the days for the highest number of absences occur with equal frequencies during a five-day work week? Test at a 5% significance level.

Day of the week	Monday	Tuesday	Wednesday	Thursday	Friday
Number of absences	15	12	9	9	15

X = the weekday a manager named as having the most absences (5 categories), $n = 60$.

1. **Assumptions.** The 60 managers are a random sample. "Absent equally" means each day has proportion $1/5 = 0.2$, so every expected count is $E = 60 \times 0.2 = 12$, which is at least 5.

2. **Hypotheses.** With $p_1, \ldots, p_5$ the population proportions for Monday through Friday,

$$H_0 : p_1 = p_2 = p_3 = p_4 = p_5 = 0.2 \qquad H_A : \text{at least one } p_i \neq 0.2$$

   In words, H_0: the absent days occur with equal frequencies (a *uniform* distribution).

3. **Test statistic.** Work down the table one row at a time:

Day	Observed O	Expected E	$O - E$	$(O-E)^2/E$
Monday	15	12.00	3.00	0.75
Tuesday	12	12.00	0.00	0.00
Wednesday	9	12.00	−3.00	0.75
Thursday	9	12.00	−3.00	0.75
Friday	15	12.00	3.00	0.75
Total	60	60.00		**3.00**

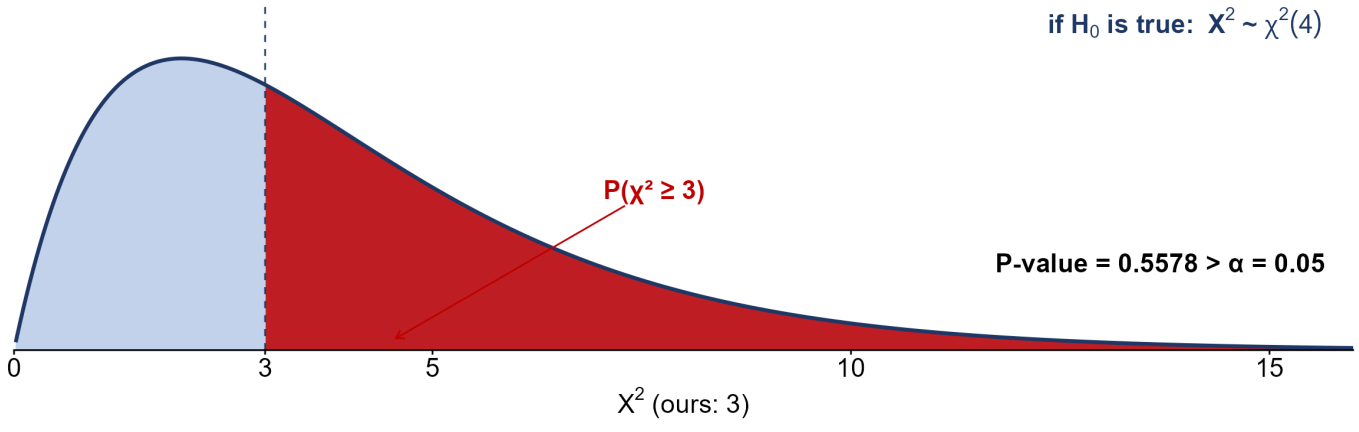
For Monday, $(15-12)^2/12 = 9/12 = 0.75$. Adding the last column,

$$X^2_{obs} = 0.75 + 0 + 0.75 + 0.75 + 0.75 = 3, \qquad df = 5 - 1 = 4$$

4. **P-value.** $X^2_{obs} = 3$ is smaller than every entry in the $df = 4$ row of the chi-square table, so the P-value is large. Software gives

$$P(\chi^2 \geq 3 \mid H_0 \text{ true}) = 0.5578$$

5. **Decision.** $0.5578 > \alpha = 0.05$: **fail to reject H_0**. The data do not give convincing evidence that the days with the most absences occur with unequal frequencies. The sample is consistent with employees being absent equally across the week, although that does not prove they are.



Notice the shape of the null distribution: a $\chi^2(4)$ curve has mean 4, so an X^2 of 3 is a perfectly ordinary value when H_0 is true. A quick sense check before using the table: **an X^2_{obs} near its df is unremarkable; one many times its df is strong evidence against H_0.**

Example: streaming services in the far West

From OpenStax Introductory Statistics 2e, Section 11.2, Example 11.3 (https://openstax.org/books/introductory-statistics-2e/pages/11-2-goodness-of-fit-test):

> One study indicates that the number of streaming services that American families have is distributed as in the table below (the percent column). A random sample of 600 families in the far western United States resulted in the data in the observed frequency column. At the 1% significance level, does it appear that the distribution "number of streaming services" of far western United States families is different from the distribution for the American population as a whole?

Number of streaming services	Percent (all U.S.)	Observed frequency (far West)
0	10	66
1	16	119
2	55	340
3	11	60
4 or more	8	15
Total	100	600

X = the number of streaming services a far-western family has (5 categories), $n = 600$.

Here the claimed distribution is not "all equal." It comes from the national percentages, so each category gets its own $p_{i,0}$.

1. **Assumptions.** A random sample of 600 families. The expected counts are $E = 600 \times 0.10 = 60$, $600 \times 0.16 = 96$, $600 \times 0.55 = 330$, $600 \times 0.11 = 66$ and $600 \times 0.08 = 48$, all at least 5.
2. **Hypotheses.** With $p_0, p_1, p_2, p_3, p_{4+}$ the proportions of far-western families with each number of services,

$$H_0 : p_0 = 0.10,\ p_1 = 0.16,\ p_2 = 0.55,\ p_3 = 0.11,\ p_{4+} = 0.08$$

$$H_A : \text{at least one of these proportions differs from its national value}$$

   In words, H_0: the far West has the same distribution as the whole country.

3. **Test statistic.**

Services	Observed O	Expected E	$O - E$	$(O-E)^2/E$
0	66	60	6	0.60
1	119	96	23	5.51
2	340	330	10	0.30
3	60	66	−6	0.55

Services	Observed	Expected		
4 or more	15	48	−33	22.69
Total	600	600		**29.65**

$$X^2_{obs} = 0.60 + 5.51 + 0.30 + 0.55 + 22.69 = 29.65, \qquad df = 5 - 1 = 4$$

(Note that $df = 4$, not $600 - 1$. The df counts categories, not families.)

4. **P-value.** The largest entry in the $df = 4$ row is 19.997 (0.0005), and $29.65 > 19.997$, so

$$P(\chi^2 \geq 29.65 \mid H_0 \text{ true}) < 0.0005$$

   Software gives 0.000006.

5. **Decision.** The P-value is below $\alpha = 0.01$: **reject** H_0. At the 1% level there is convincing evidence that the number of streaming services in far-western families has a different distribution from the country as a whole.

Where is the misfit? One term, 22.69, supplies about 77% of X^2. It is the "4 or more" category: 48 families were expected there and only 15 were observed. The next biggest, 5.51, is one service (119 observed against 96 expected). Far-western families are more likely to have exactly one service and much less likely to have four or more. Listing the large terms like this is the natural follow-up to any rejection in a chi-square test.

When an expected count is below 5

From OpenStax Introductory Statistics 2e, Section 11.2, Example 11.1 (https://openstax.org/books/introductory-statistics-2e/pages/11-2-goodness-of-fit-test):

> Absenteeism of college students from math classes is a major concern to math instructors because missing class appears to increase the drop rate. Suppose that a study was done to determine if the actual student absenteeism rate follows faculty perception. The faculty expected that a group of 100 students would miss class according to the expected column below. A random survey across all mathematics courses was then done to determine the actual number (observed) of absences in a course.

Absences per term	Expected number of students	Actual number of students
0–2	50	35
3–5	30	40
6–8	12	20
9–11	6	1
12+	2	4

Can we run the test on this table? No. The expected count for "12+" is 2, below 5. The fix is to **combine** that category with its neighbor, "9–11", into one category "9 or more", whose expected count is $6 + 2 = 8$ (and whose observed count is $1 + 4 = 5$). Combining *neighboring* categories keeps the categories meaningful. Now there are four categories, so $df = 4 - 1 = 3$.

The book stops there. Finishing the test at $\alpha = 0.05$, with H_0: student absenteeism fits the faculty perception (proportions 0.50, 0.30, 0.12, 0.08) and H_A: it does not:

Absences	Observed O	Expected E	$O - E$	$(O-E)^2/E$
0–2	35	50	−15	4.500
3–5	40	30	10	3.333
6–8	20	12	8	5.333
9 or more	5	8	−3	1.125
Total	100	100		**14.29**

$$X^2_{obs} = 4.500 + 3.333 + 5.333 + 1.125 = 14.29, \qquad df = 3$$

In the $df = 3$ row, $12.838 < 14.29 < 14.320$ (upper tails 0.005 and 0.0025), so $0.0025 < P(\chi^2 \geq 14.29 \mid H_0 \text{ true}) < 0.005$ (software: 0.0025). **Reject** H_0: actual absenteeism does not fit the faculty's perception. The biggest terms, 5.333 for 6–8 absences and 4.500 for 0–2, say students miss class *more* than the faculty think: fewer students than expected have 0–2 absences and more have 3–8.

Aside: two categories

With only two categories, a goodness-of-fit test asks the same question as a two-sided test of one proportion. The 2024 notes used Gregor Mendel's peas: Mendel's laws say a cross of two plants that each carry one yellow and one green allele gives yellow offspring with probability 0.75 and green with probability 0.25. Of his 8023 offspring, 6022 were yellow and 2001 were green.

- **Goodness of fit.** $E = 8023 \times 0.75 = 6017.25$ and $8023 \times 0.25 = 2005.75$, so

$$X^2_{obs} = \frac{(6022 - 6017.25)^2}{6017.25} + \frac{(2001 - 2005.75)^2}{2005.75} = 0.0037 + 0.0112 = 0.015, \qquad df = 2 - 1 = 1$$

- **One proportion.** $\hat{p} = 6022/8023 = 0.7506$ and

$$z_{obs} = \frac{0.7506 - 0.75}{\sqrt{0.75 \times 0.25/8023}} = 0.12$$

The two agree exactly: $X^2_{obs} = z^2_{obs}$ (keeping all the decimals, $0.1225^2 = 0.015$), and both tests give a P-value of 0.90. **A $\chi^2(1)$ variable is the square of a standard normal variable.** The data fit Mendel's model very well. The chi-square test is what lets us go beyond two categories.

The chi-square test of independence

The setup

A **test of independence** asks whether **two categorical variables**, measured on the **same** individuals from **one** random sample, are related. The data form a two-way (contingency) table with r rows and c columns.

- **The hypotheses** are stated in words:

$$H_0 : \text{the two variables are independent} \qquad H_A : \text{the two variables are not independent (they are associated)}$$

Independent means the conditional distribution of one variable is the same in every row (or column) of the other, which is what the "do the rows look alike?" comparison in the two-way tables section checked.

Where the expected counts come from. Recall from probability that events A and B are independent when $P(A \text{ and } B) = P(A)\,P(B)$. The book's Example 11.5 makes this concrete:

> Suppose, in a study of drivers who received speeding violations in the last year, and who used cell phone while driving, that 755 people were surveyed. Out of the 755, 70 had a speeding violation and 685 did not; 305 used cell phones while driving and 450 did not.

If having a speeding violation and using a cell phone while driving are independent, then

$$P(\text{violation and cell phone}) = \frac{70}{755} \times \frac{305}{755},$$

and the expected *number* of the 755 drivers in that cell is

$$E = 755 \times \frac{70}{755} \times \frac{305}{755} = \frac{70 \times 305}{755} = 28.3.$$

One 755 cancels, and what is left is the rule for every cell of a two-way table:

$$E_{ij} = \frac{(\text{row } i \text{ total}) \times (\text{column } j \text{ total})}{n}$$

- **The test statistic** adds a term for **every cell**, all $r \times c$ of them:

$$X^2_{obs} = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

- **The degrees of freedom** are

$$df = (r-1)(c-1)$$

  The row and column totals are fixed. Fill in the table one cell at a time: once $c-1$ cells of a row are known, its last cell is forced by the row total, and once $r-1$ rows are known, the last row is forced by the column totals. Only $(r-1)(c-1)$ cells are free.
- **The P-value** is again the upper tail, $P(\chi^2 \geq X^2_{obs} \mid H_0 \text{ true})$.
- **The conditions.** One random sample, each individual classified on both variables; every expected count at least 5.

In general, the observed table and its totals look like this:

	Column 1	Column 2	⋯	Column c	Row total
Row 1	O_{11}	O_{12}	⋯	O_{1c}	row 1 total
Row 2	O_{21}	O_{22}	⋯	O_{2c}	row 2 total
⋮	⋮	⋮	⋱	⋮	⋮
Row r	O_{r1}	O_{r2}	⋯	O_{rc}	row r total
Column total	column 1 total	column 2 total	⋯	column c total	n

Example: volunteers and hours

From OpenStax Introductory Statistics 2e, Section 11.3, Example 11.6 (https://openstax.org/books/introductory-statistics-2e/pages/11-3-test-of-independence): the volunteer table from the two-way tables section. A sample of 839 adult volunteers was classified by **type of volunteer** and **hours volunteered per week**. Is the number of hours volunteered independent of the type of volunteer? (No α is given, so use 0.05.)

Type of volunteer	1–3 hours	4–6 hours	7–9 hours	Row total
Community college students	111	96	48	255
Four-year college students	96	133	61	290
Nonstudents	91	150	53	294
Column total	298	379	162	**839**

Step 1: the expected counts. Each cell is (row total)(column total)/839. For community college students working 1–3 hours,

$$E_{11} = \frac{255 \times 298}{839} = 90.57.$$

The whole table:

Expected E	1–3 hours	4–6 hours	7–9 hours
Community college students	$\frac{255\times298}{839} = 90.57$	$\frac{255\times379}{839} = 115.19$	$\frac{255\times162}{839} = 49.24$
Four-year college students	$\frac{290\times298}{839} = 103.00$	$\frac{290\times379}{839} = 131.00$	$\frac{290\times162}{839} = 56.00$
Nonstudents	$\frac{294\times298}{839} = 104.42$	$\frac{294\times379}{839} = 132.81$	$\frac{294\times162}{839} = 56.77$

Check: each row of expected counts adds to its row total (for example, $90.57 + 115.19 + 49.24 = 255$), and the smallest expected count is 49.24, well above 5.

Step 2: the contributions $(O-E)^2/E$. For the top-left cell, $(111 - 90.57)^2/90.57 = 4.607$.

$(O-E)^2/E$	1–3 hours	4–6 hours	7–9 hours
Community college students	4.607	3.197	0.031
Four-year college students	0.476	0.030	0.447
Nonstudents	1.726	2.225	0.250

Now the five steps.

1. **Assumptions.** One random sample of 839 volunteers, each classified on both variables; every expected count is at least 5.
2. **Hypotheses.** H_0: the number of hours volunteered is independent of the type of volunteer. H_A: the number of hours volunteered depends on the type of volunteer.
3. **Test statistic.** Add all nine contributions:

$$X^2_{obs} = 4.607 + 3.197 + 0.031 + 0.476 + 0.030 + 0.447 + 1.726 + 2.225 + 0.250 = 12.99$$

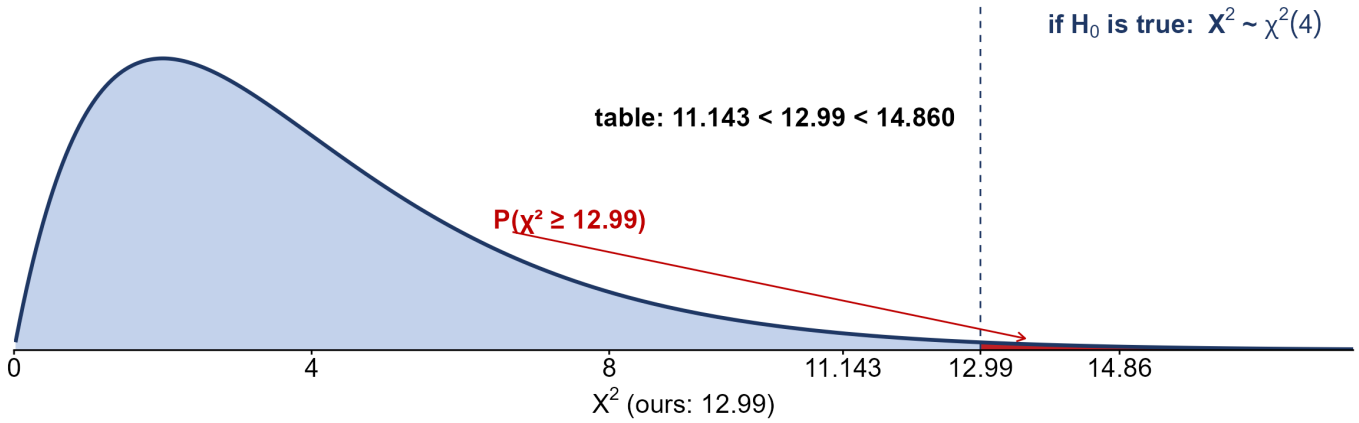
$$df = (3-1)(3-1) = 4$$

4. **P-value.** In the $df = 4$ row, $11.143 < 12.99 < 14.860$, with upper tails 0.025 and 0.005, so

$$0.005 < P(\chi^2 \geq 12.99 \mid H_0 \text{ true}) < 0.025$$

   Software gives 0.0113.

5. **Decision.** The P-value is below 0.05: **reject** H_0. There is convincing evidence that the number of hours volunteered is related to the type of volunteer.



How are they related? The largest contributions are community college students working 1–3 hours (4.607: 111 observed, 90.57 expected) and 4–6 hours (3.197: 96 observed, 115.19 expected), and nonstudents working 4–6 hours (2.225: 150 observed, 132.81 expected). Community college students volunteer fewer hours than independence predicts and nonstudents more, which is what the conditional distributions in the two-way tables section suggested.

Practice: anxiety and the need to succeed

From OpenStax Introductory Statistics 2e, Section 11.3, Example 11.7 (https://openstax.org/books/introductory-statistics-2e/pages/11-3-test-of-independence):

> De Anza College is interested in the relationship between anxiety level and the need to succeed in school. A random sample of 400 students took a test that measured anxiety level and need to succeed in school. The table shows the results. De Anza College wants to know if anxiety level and need to succeed in school are independent events.

Need to succeed	High anxiety	Med-high anxiety	Medium anxiety	Med-low anxiety	Low anxiety	Row total
High need	35	42	53	15	10	155
Medium need	18	48	63	33	31	193
Low need	4	5	11	15	17	52
Column total	57	95	127	63	58	**400**

(a) How many high-anxiety students are expected to have a high need to succeed, if the variables are independent? **(b)** How many are expected to have a low need to succeed and med-low anxiety? **(c)** Test whether anxiety level and the need to succeed are independent at $\alpha = 0.05$.

Solution.

(a) $E = \dfrac{155 \times 57}{400} = 22.09$, about 22 students. **(b)** $E = \dfrac{52 \times 63}{400} = 8.19$, about 8 students.

(c) The full table of expected counts, then the contributions:

Expected E	High anxiety	Med-high anxiety	Medium anxiety	Med-low anxiety	Low anxiety
High need	22.09	36.81	49.21	24.41	22.48
Medium need	27.50	45.84	61.28	30.40	27.98
Low need	7.41	12.35	16.51	8.19	7.54

$(O-E)^2/E$	High anxiety	Med-high anxiety	Medium anxiety	Med-low anxiety	Low anxiety
High need	7.549	0.731	0.291	3.629	6.924
Medium need	3.283	0.102	0.048	0.223	0.325
Low need	1.569	4.374	1.839	5.663	11.869

1. **Assumptions.** A random sample of 400 students; the smallest expected count is 7.41 (low need, high anxiety), so every expected count is at least 5.
2. **Hypotheses.** H_0: anxiety level and the need to succeed are independent. H_A: they are not independent.
3. **Test statistic.** The 15 contributions add to $X^2_{obs} = 48.42$, with $df = (3-1)(5-1) = 8$.
4. **P-value.** The largest entry in the $df = 8$ row is 27.868 (0.0005), and $48.42 > 27.868$, so $P(\chi^2 \geq 48.42 \mid H_0 \text{ true}) < 0.0005$.
5. **Decision. Reject H_0.** Anxiety level and the need to succeed in school are related. The largest contributions are low need with low anxiety (11.87: 17 observed, 7.54 expected) and high need with high anxiety (7.55: 35 observed, 22.09 expected): students with a higher need to succeed tend to report higher anxiety.

The chi-square test for homogeneity

The setup

A **test for homogeneity** asks whether **two or more populations** have the **same distribution** of one categorical variable. The data come from a **separate random sample from each population**, and each person is classified on that one variable. The table has a row for each population and a column for each category of the response.

$$H_0 : \text{the populations have the same distribution of the variable}$$

$$H_A : \text{the distributions are not all the same}$$

The arithmetic is identical to the test of independence:

$$E_{ij} = \frac{(\text{row } i \text{ total}) \times (\text{column } j \text{ total})}{n}, \qquad X^2_{obs} = \sum_{\text{all cells}} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \qquad df = (r-1)(c-1),$$

the P-value is the upper tail, and every expected count must be at least 5.

So what is the difference? The **design**, and therefore the question:

- **Independence:** *one* sample, and *two* variables measured on each person. The row totals and column totals are both random. Question: are the two variables related?
- **Homogeneity:** *several* samples, one per population, and *one* variable. The row totals (the sample sizes) are fixed by the researcher. Question: do the populations have the same distribution?

It is the categorical version of the jump from a one-sample test to a two-sample test: the rows are the groups we compare. With two rows, it extends the test of $p_1 = p_2$ to a response with more than two categories.

A note on the textbook: Section 11.4 of the book writes the homogeneity df as "number of columns − 1", because all of its examples compare exactly two populations. With $r = 2$ rows that is the same as $(r-1)(c-1) = (2-1)(c-1) = c-1$. Use $(r-1)(c-1)$; it works for any number of populations, and it is the formula in the book's Formula Review.

Example: living arrangements of men and women

From OpenStax Introductory Statistics 2e, Section 11.4, Example 11.8 (https://openstax.org/books/introductory-statistics-2e/pages/11-4-test-for-homogeneity):

> Do men and women college students have the same distribution of living arrangements? Use a level of significance of 0.05. Suppose that 250 randomly selected men college students and 300 randomly selected women college students were asked about their living arrangements: dormitory, apartment, with parents, other. The results are shown in the table. Do male and female college students have the same distribution of living arrangements?

	Dormitory	Apartment	With parents	Other	Row total
Men	72	84	49	45	250
Women	91	86	88	35	300
Column total	163	170	137	80	**550**

Two separate random samples (250 men, 300 women) and one variable (living arrangement), so this is a test for **homogeneity**.

Expected counts. For men in a dormitory, $E = \dfrac{250 \times 163}{550} = 74.09$. The rest:

Expected E	Dormitory	Apartment	With parents	Other
Men	74.09	77.27	62.27	36.36
Women	88.91	92.73	74.73	43.64

$(O-E)^2/E$	Dormitory	Apartment	With parents	Other
Men	0.059	0.586	2.829	2.051
Women	0.049	0.488	2.357	1.709

1. **Assumptions.** Independent random samples of men and women; the smallest expected count is 36.36, well above 5.
2. **Hypotheses.** H_0: the distribution of living arrangements is the same for men and women college students. H_A: the distributions are not the same.

3. **Test statistic.**

$$X^2_{obs} = 0.059 + 0.586 + 2.829 + 2.051 + 0.049 + 0.488 + 2.357 + 1.709 = 10.13$$

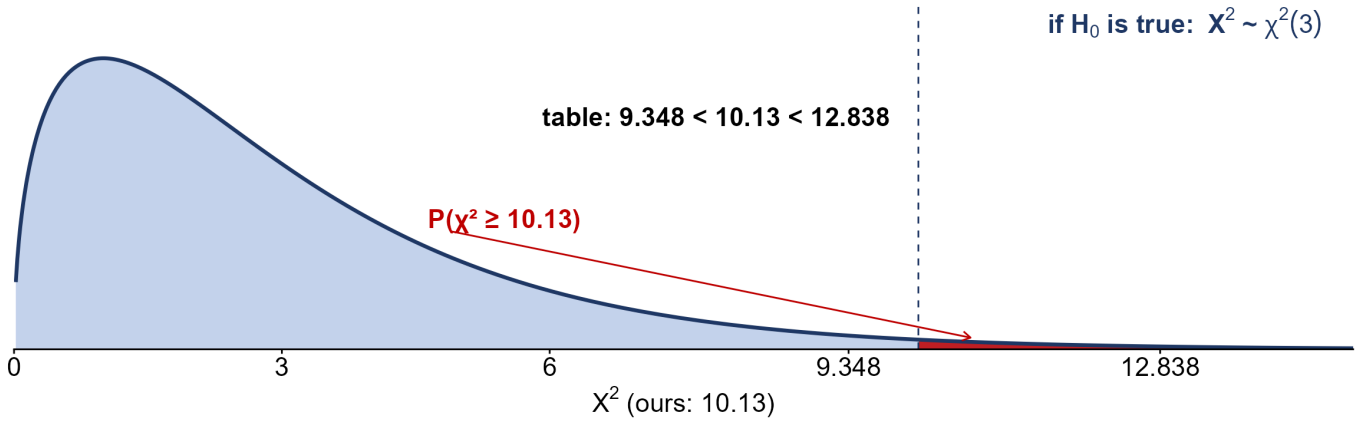
$$df = (2-1)(4-1) = 3$$

4. **P-value.** In the $df = 3$ row, $9.348 < 10.13 < 12.838$, with upper tails 0.025 and 0.005, so

$$0.005 < P(\chi^2 \geq 10.13 \mid H_0 \text{ true}) < 0.025$$

   Software gives 0.0175.

5. **Decision.** The P-value is below 0.05: **reject** H_0. There is convincing evidence that men and women college students do not have the same distribution of living arrangements.



As the book points out, the test itself only says the distributions differ, not how. The contributions show where: "with parents" (2.829 and 2.357) and "other" (2.051 and 1.709). Women are more likely to live with their parents ($88/300 = 0.293$ against $49/250 = 0.196$ of men) and less likely to have some other arrangement ($35/300 = 0.117$ against $45/250 = 0.180$).

Practice: voters before and after an earthquake

From OpenStax Introductory Statistics 2e, Section 11.4, Example 11.9 (https://openstax.org/books/introductory-statistics-2e/pages/11-4-test-for-homogeneity):

> Both before and after a recent earthquake, surveys were conducted asking voters which of the three candidates they planned on voting for in the upcoming city council election. Has there been a change since the earthquake? Use a level of significance of 0.05.

	Perez	Chung	Stevens	Row total
Before	167	128	135	430
After	214	197	225	636
Column total	381	325	360	**1066**

Solution. The two surveys are separate samples of voters taken at two times, with one variable (candidate preference): a test for homogeneity.

1. **Assumptions.** Two independent random samples (430 before, 636 after). The smallest expected count is $430 \times 325/1066 = 131.10$, so all are at least 5.
2. **Hypotheses.** H_0: the distribution of voter preferences was the same before and after the earthquake. H_A: it was not the same.
3. **Test statistic.** The expected counts and contributions are

Expected E	Perez	Chung	Stevens
Before	153.69	131.10	145.22
After	227.31	193.90	214.78

$(O-E)^2/E$	Perez	Chung	Stevens
Before	1.153	0.073	0.719
After	0.780	0.049	0.486

$$X^2_{obs} = 1.153 + 0.073 + 0.719 + 0.780 + 0.049 + 0.486 = 3.26, \qquad df = (2-1)(3-1) = 2$$

4. **P-value.** In the $df = 2$ row, $3.219 < 3.26 < 3.794$, with upper tails 0.2 and 0.15, so $0.15 < P(\chi^2 \geq 3.26 \mid H_0 \text{ true}) < 0.2$. Software gives 0.1959.
5. **Decision.** The P-value is above 0.05: **fail to reject H_0**. The data do not give convincing evidence that the distribution of voter preferences changed after the earthquake.

Which chi-square test?

(OpenStax Section 11.5 (https://openstax.org/books/introductory-statistics-2e/pages/11-5-comparison-of-the-chi-square-tests).) All three tests compute the same X^2 and use the upper tail of a chi-square distribution. Choose between them by asking **how many samples** there are and **how many variables** were measured.

Test	Design	H_0	Expected count E	df
Goodness of fit	one sample, one categorical variable	the population fits the claimed distribution ($p_i = p_{i,0}$ for every category)	$n\,p_{i,0}$	$k-1$
Independence	one sample, two categorical variables on each individual	the two variables are independent	$\frac{\text{row total} \times \text{column total}}{n}$	$(r-1)(c-1)$
Homogeneity	a separate sample from each of two or more populations, one categorical variable	the populations have the same distribution	$\frac{\text{row total} \times \text{column total}}{n}$	$(r-1)(c-1)$

All three need random sampling and every expected count at least 5, and for all three $\text{P-value} = P(\chi^2 \geq X^2_{obs} \mid H_0 \text{ true})$.

Practice: which test? For each study, name the chi-square test.

1. A random sample of 500 voters is asked both their age group (18–29, 30–49, 50–64, 65+) and their preferred news source (TV, print, online, social media). Are age group and news source related?

2. A candy company says its bags are 20% red, 20% orange, 20% yellow, 20% green and 20% blue. You count the colors in a random sample of 300 pieces.
3. Separate random samples of 200 first-year students and 200 seniors are asked how they usually get to campus (walk, bike, bus, drive). Do the two classes commute the same way?

Answers. (1) Independence: one sample, two variables measured on each voter. (2) Goodness of fit: one variable compared with a claimed distribution, with $E = 300 \times 0.2 = 60$ for each color and $df = 4$. (3) Homogeneity: two separately sampled populations, one variable, with $df = (2-1)(4-1) = 3$.

Sources

- The two-sample examples are from OpenStax *Introductory Statistics 2e*, Chapter 10: sports hours (Example 10.1), math classes (Example 10.2), online and face-to-face exams (Example 10.3), worker retention (Try It 10.2), hypnotism (Example 10.11), the softball strength class (Example 10.12), the shot-put (Example 10.13) and throwing distance (Try It 10.13). The "do not pool" quotation is from p. 513, and Cohen's d is on p. 518. The hives and electric-vehicle data are Examples 10.8 and 10.10; the 2-point claim tested in the electric-vehicle example is ours.
- The NCAA basketball scores and the couples' decision ratings come from the 2024 STAT 251 notes.
- Textbook: OpenStax *Introductory Statistics 2e*, Sections 10.1, 10.3 and 10.4, and for the categorical data and the chi-square tests, Sections 11.1–11.5.
- The volunteer table is OpenStax Example 11.6 (Table 11.15). The teens-and-phones counts are the published Pew Research Center margins used in Week 2; the "equally common" claim tested here is ours.
- The chi-square tests use OpenStax Chapter 11 examples: employee absences (Example 11.2, p. 568), streaming services (Example 11.3, p. 569), absenteeism and combining categories (Example 11.1, p. 565), speeding and cell phones (Example 11.5, p. 575), volunteers (Example 11.6, p. 575), anxiety and the need to succeed (Example 11.7, p. 578), living arrangements (Example 11.8, p. 579) and voters and the earthquake (Example 11.9, p. 580); the comparison table follows Section 11.5 (p. 582). The book gives only the two expected counts for Example 11.7 and stops at the degrees of freedom for Example 11.1; the tests that finish them are ours. Mendel's peas come from the 2024 STAT 251 notes.